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Population Biology

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Genotyping *Puccinia striiformis* f. sp. *tritici* Isolates with SSR and SP-SNP Markers Reveals Dynamics of the Wheat Stripe Rust Pathogen in the United States from 1968 to 2009 and Identifies Avirulence-Associated Markers

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ABSTRACT

Stripe rust, caused by *Puccinia striiformis* f. sp. *tritici*, is a devastating disease of wheat (*Triticum aestivum*) in the United States. The fungal pathogen can rapidly evolve, producing new virulent races infecting previously resistant cultivars and genotypes adapting to different environments. The objective of this study was to investigate the long-term population dynamics of *P. striiformis* f. sp. *tritici* in the United States. Through genotyping 1,083 isolates taken from 1968 to 2009, using 14 simple sequence repeat (SSR) markers and 92 secreted protein single nucleotide polymorphism (SP-SNP) markers, 614 and 945 genotypes were detected, respectively. In general, the two types of markers produced consistent genetic relationships among the *P. striiformis* f. sp. *tritici* populations over the 40-year period. The prior-to-2000 and the 2000-to-2009 populations were significantly different, with the latter showing higher genotypic diversity and higher heterozygosity than the earlier populations. Clustering analyses using genotypes of either SSR or SP-SNP markers revealed three molecular groups (MGs), MG1, MG2, and MG3. The prior-to-2000 and the 2000-to-2009 groups both had evidence of MG1 and MG2; however, MG3 was only found in the 2000-to-2009 population. Some of the isolates in the period of 2000 to 2009 formed individual clusters, suggesting exotic incursions. Other isolates of the same period were clustered with prior-to-2000 isolates, indicating that they were developed from the previously established populations. The data suggest the coexistence of newly introduced populations alongside established populations in the United States. Twenty SP-SNP markers were significantly associated to individual avirulence genes. These results are useful for developing more accurate monitoring systems and provide guidance for disease management.

Keywords: ecology and epidemiology, evolution, genetics, population biology, *Puccinia striiformis* f. sp. *tritici*, wheat stripe rust

Stripe rust, caused by *Puccinia striiformis* Westend. f. sp. *tritici* Erikss., is one of the most devastating diseases on wheat (*Triticum aestivum*) and continues to threaten wheat production in all major wheat-growing regions of the world (Chen 2005, 2020; Chen and Kang 2017; Wellings 2011). The global annual yield loss to wheat stripe rust was estimated to be 5.47 million tons, or U.S. $979 million (Beddow et al. 2015). Extensive efforts have been made to develop wheat cultivars resistant to stripe rust. However, resistant cultivars are frequently rendered susceptible by rapidly evolving *P. striiformis* f. sp. *tritici* populations, resulting in severe epidemics in many regions (Bayles et al. 2000; Chen 2020; Wellings et al. 2012). New virulent races of *P. striiformis* f. sp. *tritici* could be from such mechanisms as mutation, sexual recombination, and somatic hybridization (Hovmøller et al. 2011; Lei et al. 2017; Li et al. 2019b, 2020; Park and Wellings 2012; Zhao et al. 2016). New races can be spread or introduced to new wheat-growing regions in the form of asexually reproduced urediniospores, which usually serve as the initial inoculum in most epidemics (Hovmøller et al. 2010). Unconscious transmission by humans could also result in introduction of new pathogens or new races from one continent to another (Wellings and McIntosh 1990). Introduction has also been postulated to be facilitated by climate changes that allow *P. striiformis* f. sp. *tritici* to overwinter in previously unsuitable regions, e.g., high latitudes (Lyon and Broders 2017). Such emergence of new, and reemergence of previously existing, virulent races reshape the *P. striiformis* f. sp. *tritici* populations, and frequently cause severe epidemics (Chen et al. 2002; de Vallavieille-Pope et al. 2012; Hovmøller et al. 2016; Markell and Milus 2008). Therefore, investigation of the *P. striiformis* f. sp. *tritici* population dynamics, tracked over a long period, will assist in understanding the mechanisms of pathogen evolution and improving future pathogen monitoring systems.

With the help of molecular markers, studies on pathogen populations have begun to shed light on structures and dynamics of *P. striiformis* f. sp. *tritici* populations. For example, by analyses of isolates collected from 1975 to 2005 and from 2000 to 2014, Hovmøller et al. (2008, 2016) found the repeat replacement of *P. striiformis* f. sp.

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tritici populations by exotic invasion of aggressive strains in Europe. The invasion of new strains from divergent genetic lineages was also observed in worldwide *P. striiformis* f. sp. *tritici* populations (Ali et al. 2017; Sharma-Poudyal et al. 2020). Using simple sequence repeat (SSR) markers together with virulence phenotypes, Cheng and Chen (2014) determined that the *P. striiformis* f. sp. *tritici* population in the U.S. Pacific Northwest is asexually reproduced. Similarly, Cheng et al. (2016) found that grasses harbor more diverse *P. striiformis* f. sp. *tritici* populations than cereal hosts. Holtz et al. (2014) genotyped 261 isolates from different cereal crops and grasses collected from 2007 to 2012 in Alberta, Canada using 11 SSR markers and found a low genotypic diversity, but predominant genotypes changed between years. Visser et al. (2016) determined the genetic relationships of four *P. striiformis* f. sp. *tritici* races in South Africa and found that the *P. striiformis* f. sp. *tritici* population was clonal. Draz (2019) used markers that separated new *P. striiformis* f. sp. *tritici* races from old races in Egypt, and the results suggested new races from exotic introduction. Similarly, SSR markers revealed diverse genetic *P. striiformis* f. sp. *tritici* populations in Nepal, but with a low pathotype diversity (Ali et al. 2018). Also, single nucleotide polymorphisms (SNPs) have been used to study *P. striiformis* f. sp. *tritici* populations. Xia et al. (2016a) developed SNP markers based on secreted protein (SP) genes and used the SP-SNP markers to reveal population diversity and differentiation of the contemporary *P. striiformis* f. sp. *tritici* population in the United States. And using SSR markers accompanied by SNPs obtained from genome resequencing, Brar et al. (2018) revealed divergent genetic lineages in the *P. striiformis* f. sp. *tritici* population of Canada showing potential new invasions. All these studies suggested shifts in the dynamics of *P. striiformis* f. sp. *tritici* populations on both a global and a regional scale. However, most previous studies have been focused on the changes of *P. striiformis* f. sp. *tritici* populations within a short time-period. Detailed comparison of historical *P. striiformis* f. sp. *tritici* population has been limited, mainly because of a given facility's limitation for storing historical isolates. There is only one study of historical isolates of *P. striiformis* f. sp. *tritici* in the historic Stubbs *P. striiformis* f. sp. *tritici* Culture collection, which is stored and is being maintained at the Global Rust Reference Center in Denmark. Thach et al. (2016) genotyped 212 *P. striiformis* f. sp. *tritici* isolates from the Stubbs collection, representing samples from five continents taken between 1958 and 1991 using SSR markers, and identified a high diversity and recombinant population structure in China and South Asia, but a clonal population structure in northwestern Europe, East Africa, and the Mediterranean region.

In the United States, since the 1960s, *P. striiformis* f. sp. *tritici* populations have been continually characterized by their virulence (Chen et al. 2002, 2010; Line 2002; Line and Qayoum 1992; Liu et al. 2017; Wan and Chen 2014; Wan et al. 2016). They have only occasionally been characterized by using molecular markers (Chen et al. 1993; Cheng and Chen 2014; Xia et al. 2016a). It has long been noticed, based on virulence profiles, that a new *P. striiformis* f. sp. *tritici* population emerged around 2000—one that has since been the cause of severe epidemics (Chen et al. 2002). Before 2000, wheat stripe rust mainly occurred in regions west, and only rarely east, of the Rocky Mountains. Since 2000, the disease has occurred more frequently, and with growing severity, in the regions east (Chen et al. 2010; Wan and Chen 2014; Wan et al. 2016). Meanwhile, Markell and Milus (2008) observed that the post-2000 *P. striiformis* f. sp. *tritici* populations were more aggressive, and better adapted to high temperatures, than the prior-to-2000 isolates in the southern Great Plains of the United States. A similar case has also been observed in Canada, in which frequent epidemics of stripe rust has been reported (Lyon and Broders 2017).

Accompanying the exotic introduction of aggressive strains, change-of-climate patterns have also been considered to have contributed to the geographic expansion of the wheat stripe rust pathogen. As of this writing, detailed information on the dynamics of *P. striiformis* f. sp. *tritici* populations has not been available for the United States. To address this, we have recovered and examined the *P. striiformis* f. sp. *tritici* isolates collected throughout the United States from 1968 to 2009, historical information that is maintained at our U.S. Department of Agriculture, Agricultural Research Service, Stripe rust program in Pullman, Washington (Liu et al. 2017). Analyses of virulence phenotypes of these isolates using Wan and Chen (2014)'s set of 18 *Yr* single-gene differentials have shown unprecedented insights into the virulence dynamics of the historical *P. striiformis* f. sp. *tritici* populations (Liu et al. 2017). For example, a greater number of novel races, and races with a greater number of virulence genes, were detected from the 2000 to 2009 populations. From this research, it appears that several virulence genes were in existence, or were introduced into the United States, much earlier than previously estimated. The objectives of our study were to genotype the historical isolates with SSR and SP-SNP markers, use the genotypic data to investigate the genetic diversity and population dynamics of the long-term *P. striiformis* f. sp. *tritici* populations in the United States, and identify the specific molecular markers, especially those of SP-SNPs, which are associated with avirulence genes.

MATERIALS AND METHODS

***P. striiformis* f. sp. *tritici* isolates.** A total of 1,083 uredinial isolates collected from 1968 to 2009 in the United States were used in this study. These isolates have been preserved in liquid nitrogen. The criteria for selecting the isolates and the method for multiplication of urediniospores were described in Liu et al. (2017). More isolates used in this study than the 908 isolates used in the virulence study (Liu et al. 2017) were mainly because of the fact that the extra isolates had adequate original spores for DNA extraction, but were not viable for virulence tests. Similar to the previous virulence study (Liu et al. 2017), the period of 1968 to 2009 was divided into nine periods of roughly every 5 years, except the period 1998 to 1999, to facilitate comparisons of populations in different periods. The number of isolates from each period was: 92 in 1968 to 1972; 75 in 1973 to 1977; 53 in 1978 to 1982; 205 in 1983 to 1987; 89 in 1988 to 1992; 79 in 1993 to 1997; 91 in 1998 to 1999; 232 in 2000 to 2004; and 167 in 2005 to 2009 (Table 1). For more detailed information, the yearly populations were also analyzed in this study. The number of samples from each year ranged from 10 in 1973 to 64 in 2000, for an average of 33 (Supplementary Tables S1 and S2).

DNA extraction. For each isolate, 15 mg of dried urediniospores were used for extracting genomic DNA. A mixture of urediniospores and 50 µl per 0.1 mm of sterile glass beads in each tube was shaken for 2 min in a Mini-Beadbeater (BioSpec Products, Bartlesville, OK), followed by a modified cetyltrimethylammonium bromide procedure as described by Chen et al. (1993). The DNA concentration in the stock solution was quantified using a NanoDrop 1000 Spectrophotometer (Thermo Fisher Scientific, Waltham, MA). A working solution of 20 ng/µl was made by diluting the stock solution with sterilized ddH_2O and used for PCR. The quality of the DNA working solution was further checked by PCR amplification with the internal transcribed spacer (ITS) primers ITS3puc (5′-ACATCGATGAA GAACACAGT-3′; Wang et al. 2015) and ITS4B (Gardes and Bruns 1993) for all 1,083 isolates. The PCR amplicons were checked on 0.8% agarose gel. Note that the 304-base pair (bp) target bands should have been clearly present for each isolate; if they were not, the stock solution was repurified and/or the working solution was remade and checked before genotyping with SSR and SP-SNP markers.

SSR genotyping. Fourteen SSR primers developed for *P. striiformis* f. sp. *tritici*, including RJ8, RJ18, RJ20, and RJ21 (Enjalbert et al. 2002), CPS02, CPS04, CPS08, and CPS13 (Chen et al. 2009), and Pstp002, Pstp033, Pstp004, Pstp029, Pstp006, and Pstp033 (Cheng et al. 2012), were used to characterize the *P. striiformis* f. sp. *tritici* isolates in this study. The PCR reactions contained 1.2 µl of 10× PCR buffer, 0.48 µl of 25-mM $MgCl_2$, 0.96 µl of 2.5-mM dNTP, 0.03 µl of 10 µM primer FM13, 0.6 µl of 10 µM primer R, 25 ng

of genomic DNA, 5 U/µl of *Taq* polymerase, and 4.56 µl of ddH_2O. PCR reaction conditions were 95°C for 5 min, followed by 42 cycles consisting of 1 min at 95°C, 1 min at 50°C to 60°C depending upon primers, and 2 min at 72°C, with a final extension step of 10 min at 72°C. PCR products were analyzed in an Applied Biosystems 3730xl DNA Analyzer (Thermo Fisher Scientific, Waltham, MA). The sizes of the PCR products were estimated using an internal DNA size standard method (GeneScan 500 LIZ dye Size Standard; Thermo Fisher Scientific). Allele sizes in base pairs were determined using the software GeneMarker 4.0 (SoftGenetics, State College, PA).

SNP genotyping. In total, 92 SP-SNP markers (Xia et al. 2016a, b) were used for SNP genotyping. The PCR conditions and detection of SNP described by Xia et al. (2016a, b) were followed, and genotyping was conducted on a MassARRAY iPlex platform (SEQUENOM, San Diego, CA). The SNP alleles were called and manually checked using the program MassARRAY TypeAnalyzer (Agena Bioscience, San Diego, CA).

Data analyses. The *P. striiformis* f. sp. *tritici* isolates were defined as nine populations with a time-period each of roughly 5 years as one population. The R-based program POPPR v.2.8.2 (Kamvar et al. 2014) was used to determine the number of multilocus genotypes (MLGs), genotypic diversity (Stoddart and Taylor 1988), and Nei's unbiased gene diversity (Nei 1978) across the nine populations. The genotypic diversity across all time periods were estimated based on both the number of MLGs/number of the isolates (*g*/*N*) and the Stoddart and Taylor's index/numbers of MLGs (*G*/*g*) (Grünwald et al. 2003).

To test for the mode of reproduction of *P. striiformis* f. sp. *tritici* population, the observed linkage among loci against expected distributions from permutation were evaluated using the standard index of association (*rBarD*) in the R-based program POPPR v.2.8.2 with all MLGs in each population. The null hypothesis *rBarD* = 0 was tested using 1,000 permutations. If the observed *rBarD* value was located outside the distribution of the randomized dataset at $P < 0.001$, the population was considered clonal (Kamvar et al. 2014).

To identify putative clusters of genetically related isolates, the hierarchical clustering analysis using dissimilarities and the "ward.D2" method with the "hclust" function in the program R stats v.3.6.2 (Murtagh and Legendre 2014), and the multivariate discriminant analysis of principal components (DAPC) using the "dapc" function in the Adegenet package of the R program (Jombart 2008; Jombart et al. 2010), were performed. The parameters for hierarchical cluster analysis were as described in Sharma-Poudyal et al. (2020). The analysis of molecular variance (AMOVA) using the "poppr.amova" function in the program POPPR was used to calculate the population genetic variance using 999 permutations. To detect the population differentiation, the corrected standardized fixation index (Hedrick 2005; Meirmans and Hedrick 2011) between the populations was calculated using the program GenAlEx 6.5 (Peakall and Smouse 2012).

To determine the correlations between the SSR and SP-SNP data, and between molecular data sets obtained in this study and the virulence data in Liu et al. (2017), the distance matrix was generated for each year-population with each of the three data sets with the program GenAlEx v.6.5 (Peakall and Smouse 2012). The distance values between isolates were divided by the mean value of the population for normalization, and then regression analysis was used to calculate the correlation coefficient between each two datasets. Correlation analysis was conducted between individual molecular marker locus and avirulence genes to identify marker loci significantly associated with individual avirulence loci, using the program GAPIT v.3 (Wang and Zhang 2018) and the procedure from Xia et al. (2016b).

RESULTS

Multilocus genotypes and population diversity. A total of 614 and 944 MLGs was detected from the 1,083 *P. striiformis* f. sp. *tritici*

TABLE 1. Numbers of *Puccinia striiformis* f. sp. *tritici* isolates, multilocus genotypes (MLGs), diversity indices, heterozygous locus frequency, number and frequency of isolates in each molecular group (MG), and probability (*P*) of being sexually produced in different periods of 1968 to 2009[a]

Marker type and period	Number of isolates (*N*)	Number of MLGs (*g*)	*g*/*N*	Stoddart and Taylor's index (*G*)	*G*/*g*	No. (%) of private MLGs	Heterozygosity (expected)[b]	No. of isolates and frequency (%) in each MG[c]			*rBarD*[d]	*P* value[d]
								MG1	MG2	MG3		
SSRs												
1968 to 1972	92	80	0.87	62.2	0.78	43 (53.7)	0.46	36 (39.1)	47 (51.1)	9 (9.8)	0.14	<0.001
1973 to 1977	75	64	0.85	55.7	0.87	23 (35.9)	0.38	15 (20.0)	59 (78.7)	1 (1.3)	0.21	<0.001
1978 to 1982	53	47	0.89	43.2	0.92	14 (29.8)	0.35	5 (9.4)	48 (90.6)	0 (0.0)	0.17	<0.001
1983 to 1987	205	150	0.73	104.5	0.70	67 (44.7)	0.36	24 (11.7)	178 (86.4)	3 (1.5)	0.18	<0.001
1988 to 1992	89	70	0.79	53.9	0.77	19 (27.1)	0.37	12 (13.5)	75 (84.3)	2 (2.2)	0.20	<0.001
1993 to 1997	79	65	0.82	55.2	0.85	28 (43.1)	0.35	0 (0.0)	69 (87.3)	10 (12.7)	0.16	<0.001
1998 to 1999	91	82	0.90	72.0	0.88	38 (46.3)	0.37	5 (5.5)	76 (83.5)	10 (11.0)	0.16	<0.001
Prior 2000	*684*	*375*	*0.55*	*178.2*	*0.48*	*232 (61.9)*	*0.38*	*97 (14.3)*	*552 (80.6)*	*35 (5.1)*	*0.18*	*<0.001*
2000 to 2004	232	189	0.81	162.1	0.86	119 (63)	0.44	21 (9.9)	85 (36.6)	126 (54.3)	0.14	<0.001
2005 to 2009	167	121	0.72	97.9	0.81	65 (53.7)	0.43	24 (14.4)	34 (20.3)	109 (65.3)	0.11	<0.001
2000 to 2009	*399*	*299*	*0.75*	*242.0*	*0.81*	*184 (61.5)*	*0.44*	*45 (11.3)*	*119 (30.1)*	*235 (58.6)*	*0.13*	*<0.001*
Overall mean	1,083	614	0.57	289.9	0.47	416 (67.8)	0.43	142 (13.1)	671 (61.9)	270 (25.0)	0.17	<0.001
SP-SNPs												
1968 to 1972	92	70	0.76	59.6	0.85	49 (70.0)	0.33	38 (41.3)	54 (58.7)	0 (0.0)	0.17	<0.001
1973 to 1977	75	68	0.91	63.2	0.93	54 (79.4)	0.30	17 (22.7)	58 (77.3)	0 (0.0)	0.25	<0.001
1978 to 1982	53	49	0.92	41.9	0.86	41 (83.7)	0.28	5 (9.4)	48 (90.6)	0 (0.0)	0.24	<0.001
1983 to 1987	205	179	0.87	153.9	0.87	149 (83.7)	0.30	24 (11.7)	181 (88.3)	0 (0.0)	0.20	<0.001
1988 to 1992	89	87	0.98	85.2	0.98	80 (92)	0.31	12 (13.5)	77 (86.5)	0 (0.0)	0.18	<0.001
1993 to 1997	79	79	1.00	79.0	1.00	78 (98.7)	0.29	0 (0.0)	79 (100.0)	0 (0.0)	0.09	<0.001
1998 to 1999	91	90	0.99	89.0	0.99	88 (97.8)	0.30	3 (2.2)	88 (97.8)	0 (0.0)	0.13	<0.001
Prior 2000	*684*	*599*	*0.88*	*505.0*	*0.84*	*539 (89.8)*	*0.31*	*99 (14.5)*	*585 (85.5)*	*0 (0.0)*	*0.17*	*<0.001*
2000 to 2004	232	228	0.98	222.4	0.98	209 (91.7)	0.36	8 (3.4)	105 (45.3)	119 (51.3)	0.15	<0.001
2005 to 2009	167	135	0.81	118.7	0.87	90 (66.2)	0.38	34 (20.5)	44 (26.5)	89 (53.6)	0.14	<0.001
2000 to 2009	*399*	*346*	*0.87*	*304.0*	*0.88*	*299 (86.4)*	*0.37*	*42 (8.8)*	*149 (40.1)*	*208 (51.1)*	*0.14*	*<0.001*
Overall mean	1,083	944	0.87	808.2	0.86	838 (88.8)	0.35	141 (12.1)	734 (69.1)	208 (18.8)	0.14	<0.001

[a] SSRs = simple sequence repeat markers; SP-SNPs = secreted protein gene-derived single nucleotide polymorphism marker loci; number of isolates (*N*), number of MLGs (*g*), Stoddart and Taylor's genotypic diversity index (*G*).
[b] Nei's unbiased gene diversity over 14 SSR or 92 SP-SNP loci (Nei 1978).
[c] MGs are molecular groups identified using phylogenetic dendrograms. MG1 = molecular group 1, MG2 = molecular group 2, and MG3 = molecular group 3.
[d] *P* value of the standardized index of association, *rBarD*. $P < 0.001$ indicates significant disequilibrium because of linkage among loci, indicating populations are clonal.

isolates using the 14 SSR and 92 SP-SNP loci, respectively (Table 1; Supplementary Tables S1 and S2). The larger number of MLGs detected by using SP-SNP rather than SSR loci was because of the larger number of SP-SNP loci used in this study. The MLG diversities of different time periods were indicated by both the ratio (*g*/*N*) of the number of MLGs (*g*) over the number of isolates (*N*) and the ratio (*G*/*g*) of the Stoddart and Taylor's genotypic diversity index (*G*) over the number of MLGs (*G*/*g*). The ratio of *g*/*N* was 0.55 for the prior-to-2000 population, but 0.75 for the 2000-to-2009 population using the SSR data. When using the SP-SNP data, the ratios of *g*/*N* were 0.88 for the prior-to-2000 population and 0.87 for the 2000-to-2009 population. There was a higher genotypic diversity measured by the *G*/*g* value for the 2000-to-2009 population (0.88) than the prior-to-2000 population (0.84) based on the SP-SNP markers, and the difference was especially obvious based on the SSR data (0.48 for the prior-to-2000 population and 0.81 for the post-2000 population).

The high genetic diversity was also reflected by the high percentage of private MLGs in each time-period population. For each roughly 5-year period population, a large number of private MLGs, present in one period but absent in others, was detected. This ranged from 27.1% for the period of 1988 to 1992, to 63.0% during the period of 2000 to 2004, based on the SSR markers; and ranged from 66.2% during the period of 2005 to 2009, to 98.7% during the period of 1993 to 1997, based on the SP-SNP markers (Table 1). The percentages of private MLGs were similar between the prior-to-2000 population (61.9%) and the 2000-to-2009 population (61.5%), based on the SSR (89.8%) and the SP-SNP markers (86.4%).

The expected heterozygosity that also represents gene diversity was estimated for each roughly 5-year-period population. The average heterozygosity for each period ranged from 0.35 (1978 to 1982, and 1993 to 1997) to 0.46 (1968 to 1972) based on the SSR markers and from 0.28 (1978 to 1982) to 0.38 (2005 to 2009) based on the SP-SNP markers (Table 1). Of the nine periods, the 1968-to-1972, 2000-to-2004, and 2005-to-2009 populations showed higher heterozygosity values than the other periods, based on both the SSR and SP-SNP markers. Overall, the 2000-to-2009 population had higher heterozygosity than the prior-to-2000 population. Before 2000, the *P. striiformis* f. sp. *tritici* populations had the highest heterozygosity during the period of 1968 to 1972, but the population changed to a more homozygous one until the year 2000. Heterozygosity was then relatively stable from 2000 to 2009.

The reproduction mode was determined by the *rBarD* value and its *P* value for each time-period population and the overall population (Table 1). With all time-period populations and the overall population, the hypothesis of "no linkage among markers" was rejected ($P_{\mathrm{rBard}} < 0.001$), supporting a clonal mode of reproduction in the *P. striiformis* f. sp. *tritici* populations. This was also consistent between the different genotyping approaches.

Molecular groups. The 1,083 isolates were optimally clustered into three molecular groups (MGs), based on either the SSR or the SP-SNP data, using hierarchical cluster analysis (Fig. 1A1 and B1; Table 1; Supplementary Fig. S1). Similarly, the DAPC results based on either SSR or SP-SNP data also optimally revealed three MGs (Supplementary Fig. S1, A and B). Using the SSR data, the *K* values of 3 and 4 were indicated by the elbows in the curve based on the Bayesian information criterion (Supplementary Fig. S1 A1); however, only $K = 3$ was determined as the best clustering number, as there were obvious overlaps in the scatter plot when $K = 4$ was used

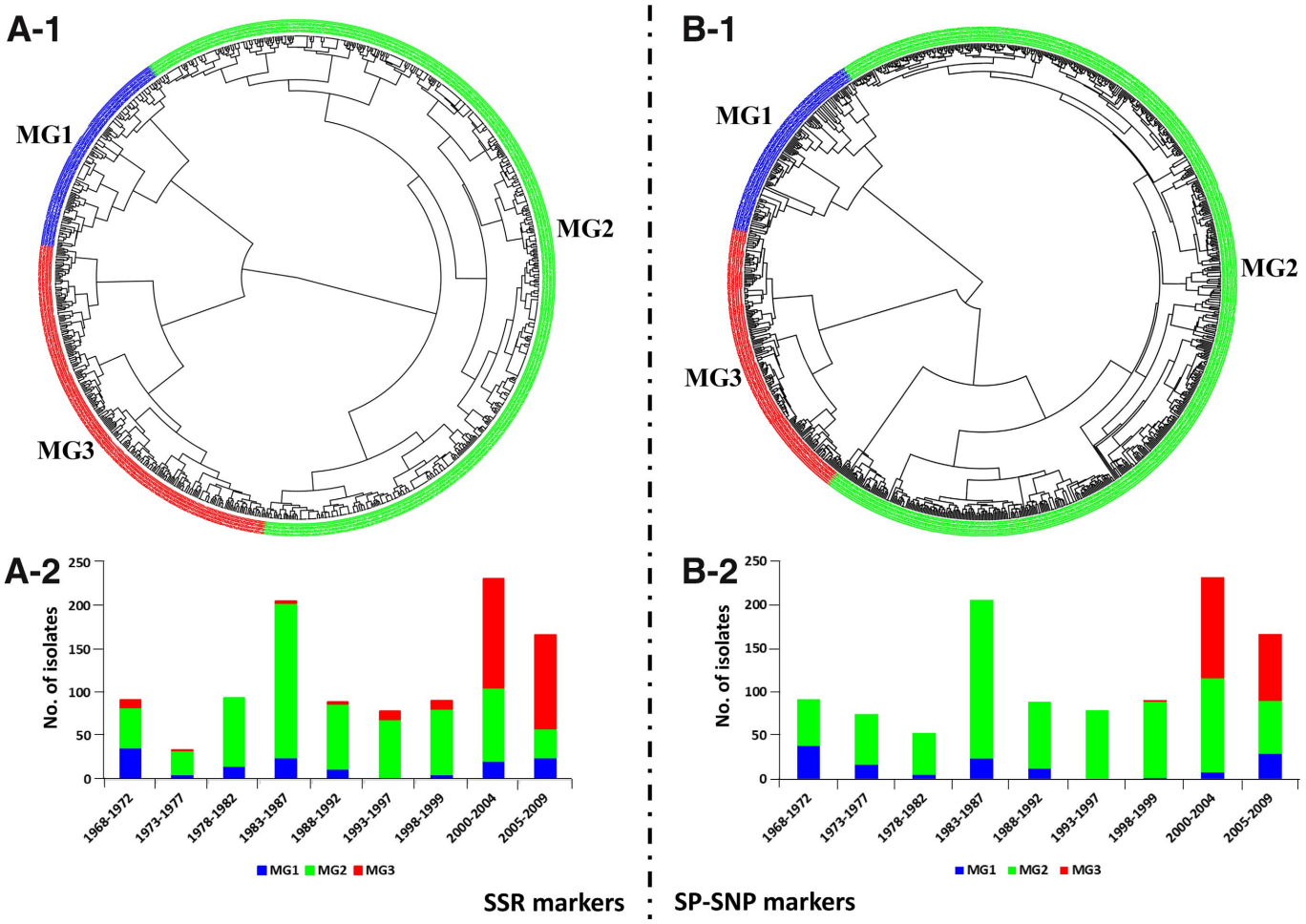

Fig. 1. Dendrograms of *Puccinia striiformis* f. sp. *tritici* populations constructed based on dissimilarities assessed with **A1,** 14 simple sequence repeat markers (SSRs) and **B1,** 92 secreted protein gene-derived single nucleotide polymorphism markers (SP-SNPs), using hierarchical cluster analysis. Shown are three molecular groups (MGs) and histograms of the MG distributions in each roughly 5-year period (**A2,** SSR; **B2,** SP-SNP).

(Supplementary Fig. S1A2 and A3). For the SP-SNP data, the optimal clustering number was only $K = 3$ (Supplementary Fig. S1B), which was consistent with the result from SSR markers and thus, the respective scatter plot was not shown. Therefore, the dynamics of the *P. striiformis* f. sp. *tritici* population was analyzed using three MGs.

The dynamics of the three MGs in the *P. striiformis* f. sp. *tritici* populations generated by the SSR and SP-SNP data were generally similar (Fig. 1A2 and B2). Counting for 13.3% of the total isolates based on the SSR data and 12.5% based on the SNP data, MG1 appeared in 1968 to 1982, gradually disappeared from 1993 to 1997, and then reemerged from 1998 to 2009. MG2 was the largest group, counting for 61.8% (based on the SSR data) or 69.5% (based on the SP-SNP data) of the total isolates. This group existed in 1968 to 1972 with a relatively low percentage (51.1 and 58.7% based on SSR and SNP markers, respectively) and kept increasing until 1978 to 1982 (90.6% based on both SSR and SNP markers). This MG slightly decreased but still at a high percentage (>80%) from 1983 to 1999; then it suddenly reduced to 36.6 and 42.4% during 2000 to 2004 and even to 19.8 and 36.5% during 2005 to 2009, based on SSR and SNP markers, respectively. MG3 was the second largest group, counting for 24.9% (based on the SSR data) or 18.0% (based on the SP-SNP data) of the 1,083 isolates (Fig. 1A1 and B1). Based on the SSR marker data, MG3, consisting of 35 prior-to-2000 isolates and 235 2000-to-2009 isolates, was detected in the 1968-to-1972 period, but disappeared until reemergence during the 1983-to-1987 period, suddenly increasing to a much higher percentage later, from 12.9% prior to 2000, to 87.1% for the 2000-to-2009 period (Fig. 1A2). MG3 was slightly different when the SP-SNP data were used. MG3 contained only two isolates (1.03%) prior to 2000, and 193 (98.97%) during the 2000-to-2009 period (Supplementary Table S2). MG3 was absent from 1968 to 1997, emerged in 1998 to 1999, and then fully developed in the predominant group during the 2000-to-2009 period (Fig. 1B2).

The minimum spanning network trees of the relationships between individual MLGs are shown in Figure 2, with one including all isolates of the three MGs (Fig. 2A) and one including only isolates of MG3 across all time periods (Fig. 2B), based on SSR markers. The three MGs were generally distinct from each other, although some individual MLGs in different MGs were close to each other (Fig. 2A). In general, MG1 was more closely related to MG3 than to MG2. Both MG2 and MG3 were more diverse than MG1. In MG3, most of the MLGs of the 2000-to-2009 period, especially these circled by the broken curves, were genetically distant from those of the prior-to-2000 period; but some MLGs of the 2000-to-2009 period were closely related to the majority of prior-to-2000 MLGs in MG3 (Fig. 2B). Because the 2000-to-2009 populations had both of the old groups (MG1 and MG2) as well as the new group (MG3), the *P. striiformis* f. sp. *tritici* population became more diverse.

Genetic relationships of *P. striiformis* f. sp. *tritici* populations on a 5-year period and on a yearly basis. The phylogenetic dendrograms and DAPC scatter plots at the 5-year periods are shown in Figure 3. In the phylogenetic dendrograms based on either the SSR (Fig. 3A1) or SNP (Fig. 3B1) markers, the prior-to-2000 populations were clearly separated from the 2000-to-2009 populations by a 100% bootstrap value. Moreover, the 1968-to-1972 population was also differentiated clearly from the other prior-to-2000 populations by a 96.4% bootstrap value on SSR markers and 78.4% on SNP markers. Similarly, in the DAPC scatter plots, the 2000-to-2009 isolates, indicated by red and dark-red dots, were obviously differentiated from the prior 2000 isolates, and most of the isolates of the 1968-to-1972 period, indicated by blue dots, were also distributed in different coordination from other time-period isolates based on either SSR (Fig. 3A2) or SP-SNP (Fig. 3B2) markers; this is especially notable for the SP-SNP markers. Based on the density plots using either the SSR data or SP-SNP data (Supplementary Fig. S2), the 2000-to-2009 populations had isolates of both types represented by the red and dark red colors, indicating that the *P. striiformis* f. sp. *tritici* population significantly differentiated around the year 2000.

To detect more detailed genetic relationships of the *P. striiformis* f. sp. *tritici* populations from 1968 to 2009, phylogenetic analysis on a

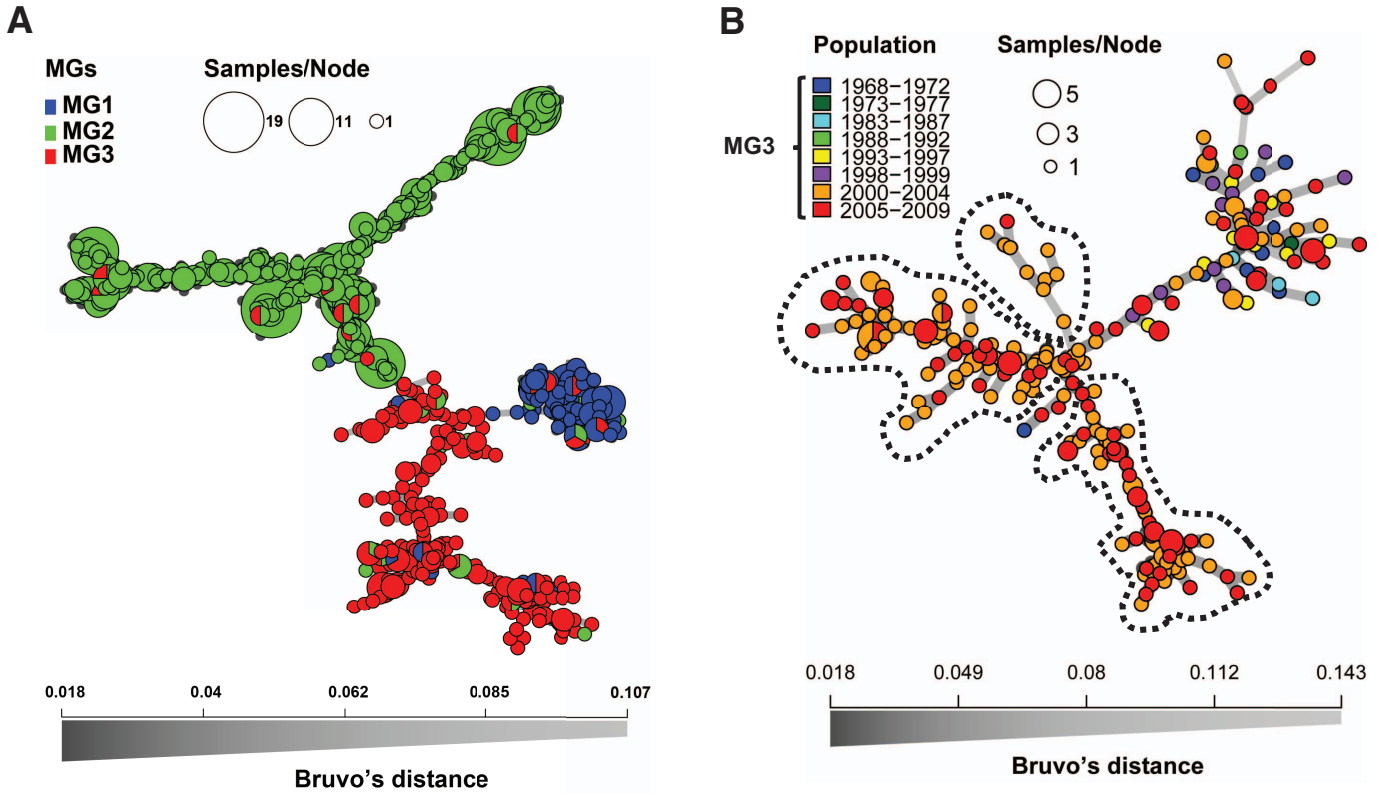

Fig. 2. Minimum spanning networks of *Puccinia striiformis* f. sp. *tritici* populations based on Bruvo's distance showing the genetic relationships of **A,** isolates in the three molecular groups (MGs; MG1, MG2, and MG3) from 1968 to 2009; and **B,** isolates in MG3 across the nine roughly 5-year periods assessed with the simple sequence repeat markers.

yearly basis was also conducted using the SSR and SP-SNP data separately (Fig. 4). Both data revealed three branches in the phylogenetic trees (Fig. 4A for SSR; Fig. 4B for SP-SNP). The 2000-to-2009 populations were clearly separated from the prior-to-2000 populations. The prior-to-2000 populations were further separated into two groups: one for the 1968, 1971, and 1972 populations, and another, bigger group to include all the other "prior to 2000" years. There were slight differences among the three branches in the phylogenetic trees based on SSR and SNP markers. The separation of the group of 1968, 1971, and 1972 from the other years was more distinct by the SP-SNP markers (Fig. 4B) than by the SSR markers (Fig. 4A). Interestingly, in the branch including 1970 and 1973 to 1999, there were two similar subbranches: one included 1973, 1985, 1991, and 1992 based on the SSR markers and 1973, 1991, and 1992 based on the SNP markers; and the other consisted of the remaining years in this branch. Similarly, in the 2000-to-2009 branch, the populations of 2000, 2001, 2002, and 2006 were clustered in a subbranch based on both SSR markers and SNP markers. Overall, the genetic relationships of the populations based on the nine time-periods and 33 years were coincidently revealed by both SSR and SP-SNP data.

Genetic variation and population differentiation. The isolates were grouped according to three types of time spans, including two time periods (prior to 2000 and 2000 to 2009), nine time periods (1968 to 1972, 1973 to 1977, 1978 to 1982, 1983 to 1987, 1988 to 1992, 1993 to 1997, 1998 to 1999, 2000 to 2004, and 2005 to 2009), and 33 individual years. The genetic variation of the *P. striiformis* f. sp. *tritici* populations from different types of time spans was partitioned by AMOVA using SSR and SP-SNP data separately (Table 2). The highest variation was within the years, at 72.22% based on the SSR data and 75.36% based on the SP-SNP data; this was followed by between the two periods (prior-to-2000 and 2000-to-2009) at 22.97 and 15.37% based on SSR and SP-SNP data, respectively. The variation among the rough 5-year periods between the prior-to-2000 and 2000-to-2009 periods was only 1.38% based on the SSR data and 4.52% based on the SP-SNP data. Similarly, the variation among years within a "5-year" period was also small, at 3.43% based on the SSR data and 4.75% based on the SP-SNP data.

The corrected standardized fixation index ($G'[ST]$) was used to estimate the genetic differentiation among the populations of different time periods. The $G'(ST)$ estimations and P values of the pairwise comparisons among the nine "5-year" period populations are shown in Table 3. The differentiations were significant between all periods except for 1973 to 1977 and 1978 to 1982; 1973 to 1977 and 1998 to 1999; 1978 to 1982 and 1983 to 1987; 1978 to 1982 and 1988 to 1992; 1978 to 1982 and 1998 to 1999; 1983 to 1987 and 1988 to 1992; 1983 to 1987 and 1998 to 1999; and 1988 to 1992 and 1998 to 1999, based on the SSR data. Based on the SP-SNP data, the differentiations were all significant, except 1973 to 1977 and 1978 to 1982; 1978 to 1982 and 1983 to 1987; 1988 to 1992, and 1998 to 1999; and 1993 to 1997 and 1998 to 1999. Both the SSR and SP-SNP data revealed significant differentiations between 1968 to 1973 and the other "5-year" periods and between prior-to-2000 and 2000-to-2009 populations.

Correlation between molecular genotypes and virulence phenotypes. The SSR and SP-SNP genotypes were significantly correlated when isolates from the overall 1968 to 2009 populations were

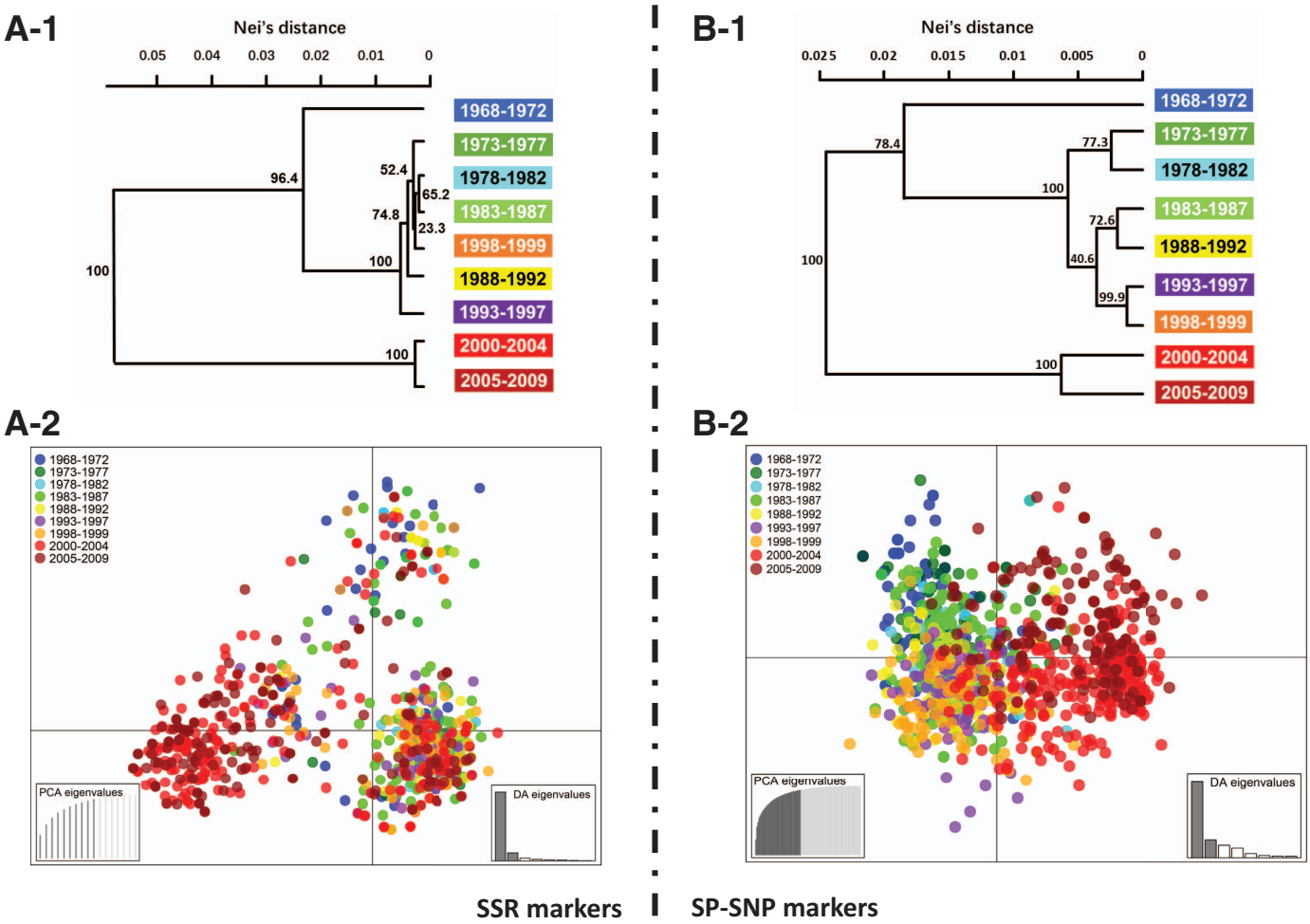

Fig. 3. Unweighted pair group arithmetic mean dendrograms showing the genetic relationships of the nine roughly 5-year period populations of *Puccinia striiformis* f. sp. *tritici* based on the Nei's genetic distances assessed with the **A1,** 14 simple sequence repeat (SSR) and **B1,** 92 secreted protein gene-derived single nucleotide polymorphism (SP-SNP) markers. Shown also is discriminant analysis of principal components of the 1,083 isolates from nine 5-year periods assessed with the **A2,** SSR and **B2,** SP-SNP markers. DA, discriminant analysis; PCA, principal component analysis.

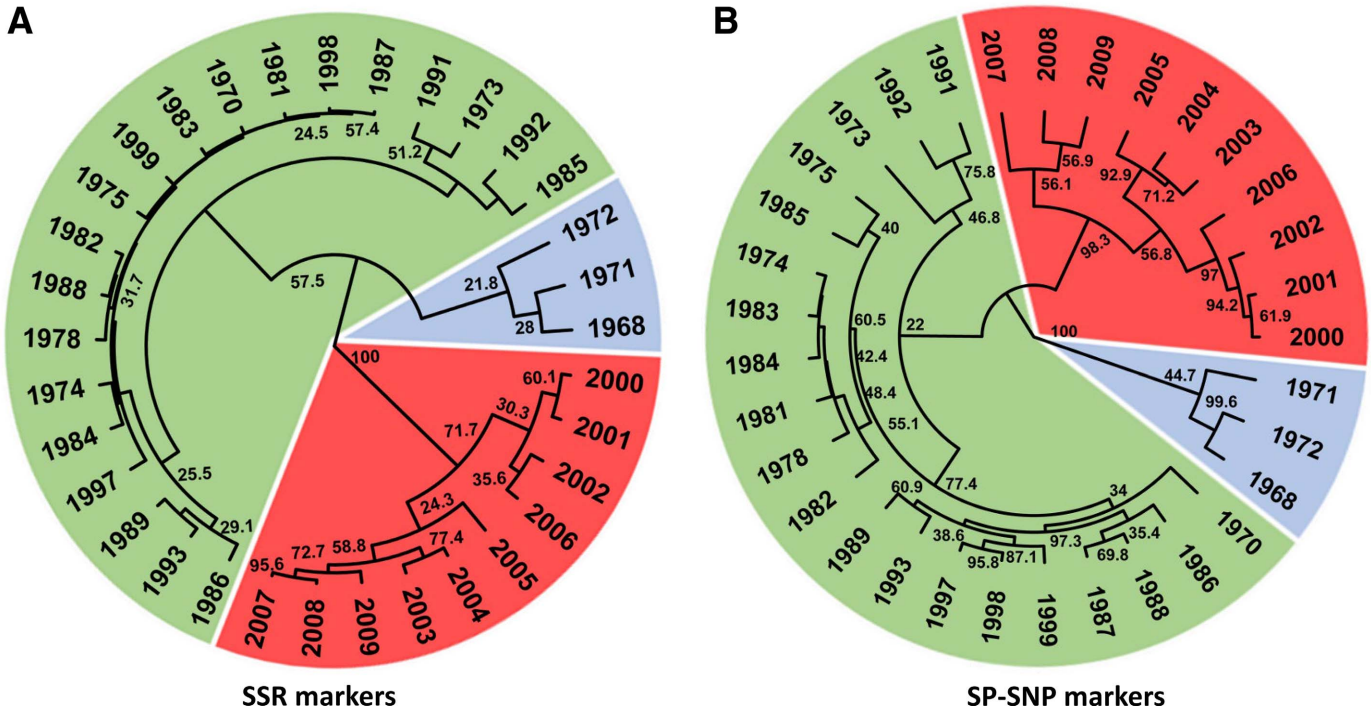

Fig. 4. Unweighted pair group arithmetic means dendrograms showing the genetic relationships of the yearly populations of *Puccinia striiformis* f. sp. *tritici* from 1968 to 2009 based on the Nei's genetic distances assessed with the **A,** 14 simple sequence repeat (SSR) and **B,** 92 secreted protein gene-derived single nucleotide polymorphism (SP-SNP) markers.

TABLE 2. Analysis of genetic variance for populations of *Puccinia striiformis* f. sp. *tritici* from 1968 to 2009 in the United States based on time periods for genetic differentiation at 14 simple sequence repeat markers (SSRs) and 92 secreted protein gene-derived single nucleotide polymorphism marker loci (SP-SNPs)

Source of variation	df	Sum of squares	Mean squares	Percentage of variation (%)	*P* value[a]
SSRs					
Among two periods (prior-2000 and 2000 to 2009)	1	465.40	465.40	22.97	0.035
Among nine periods within two periods	7	92.23	13.18	1.38	0.044
Among years within nine periods	24	168.10	7.00	3.43	0.001
Within years	1,051	2,934.91	2.79	72.22	0.001
SP-SNPs					
Among two periods (prior-2000 and 2000 to 2009)	1	1,206.83	1,206.83	15.37	0.021
Among nine periods within two periods	7	700.51	100.07	4.52	0.001
Among years within nine periods	24	743.57	30.98	4.75	0.001
Within years	1,051	10,854.41	10.33	75.36	0.001

[a] Significance tests were performed using 999 permutations.

TABLE 3. Population differentiation of *Puccinia striiformis* f. sp. *tritici* in different periods indicated by a corrected standardized fixation index ($G'[ST]$) based on 14 simple sequence repeat markers (SSRs) and 92 secreted protein gene-derived single nucleotide polymorphism marker loci (SP-SNPs)[a]

Marker type and periods	1968 to 1972	1973 to 1977	1978 to 1982	1983 to 1987	1988 to 1992	1993 to 1997	1998 to 1999	2000 to 2004	2005 to 2009
SSRs									
1968 to 1972		**0.001**	**0.001**	**0.001**	**0.001**	**0.001**	**0.001**	**0.001**	**0.001**
1973 to 1977	0.079		0.288	0.234	**0.042**	**0.003**	0.441	**0.001**	**0.001**
1978 to 1982	0.100	0.002		0.699	0.368	**0.060**	0.536	**0.001**	**0.001**
1983 to 1987	0.097	0.002	–0.003		0.106	**0.014**	0.197	**0.001**	**0.001**
1988 to 1992	0.082	0.012	0.001	0.004		**0.018**	0.064	**0.001**	**0.001**
1993 to 1997	0.133	0.022	0.010	0.011	0.012		**0.012**	**0.001**	**0.001**
1998 to 1999	0.083	–0.001	–0.002	0.002	0.008	0.015		**0.001**	**0.001**
2000 to 2004	0.144	0.212	0.235	0.226	0.205	0.208	0.188		**0.007**
2005 to 2009	0.200	0.308	0.339	0.329	0.310	0.316	0.284	0.017	
SP-SNPs									
1968 to 1972		**0.007**	**0.001**	**0.001**	**0.001**	**0.001**	**0.001**	**0.001**	**0.001**
1973 to 1977	0.045		0.068	**0.001**	**0.001**	**0.001**	**0.001**	**0.001**	**0.001**
1978 to 1982	0.112	0.014		0.356	**0.013**	**0.001**	**0.003**	**0.001**	**0.001**
1983 to 1987	0.122	0.031	0.001		**0.001**	**0.001**	**0.001**	**0.001**	**0.001**
1988 to 1992	0.171	0.057	0.020	0.021		**0.016**	0.233	**0.001**	**0.001**
1993 to 1997	0.248	0.115	0.054	0.052	0.012		0.088	**0.001**	**0.001**
1998 to 1999	0.211	0.077	0.027	0.029	0.002	0.005		**0.001**	**0.001**
2000 to 2004	0.169	0.085	0.054	0.058	0.107	0.146	0.106		**0.001**
2005 to 2009	0.074	0.052	0.068	0.079	0.120	0.172	0.137	0.052	

[a] $G'(ST)$ values are shown below the diagonal. *P* values based on 999 permutations are shown above the diagonal. *P* values in bold indicate significant differentiations.

analyzed ($r = 0.813$, $P < 0.0001$) and the isolates of prior-to-2000 analyzed ($r = 0.838$, $P < 0.0001$), but the correlation was not significant for the isolates of 2000 to 2009 ($r = 0.374$, $P = 0.287$; Table 4). The correlation coefficient between the SSR genotypes and virulence phenotypes was 0.275 for the overall (1968-to-2009) populations, 0.274 for the prior-to-2000 populations, and 0.424 for the 2000-to-2009 populations, but none of the values were significant ($P = 0.121$ to 0.222). The genotypes accessed with the 92 SP-SNP markers were significantly correlated with the virulence genotypes when all isolates from the overall 1968-to-2009 populations were analyzed ($r = 0.356$, $P = 0.042$); the correlation was much higher when the 2000-to-2009 data were analyzed ($r = 0.746$, $P = 0.042$); however, the correlation was not significant for the prior-to-2000 populations ($r = 0.331$, $P = 0.123$).

To identify individual markers associated to individual avirulence loci, correlation coefficients were obtained for every pair of the SP-SNP and SSR loci with the virulence loci in the *Yr* single-gene line genotypes used to differentiate races of *P. striiformis* f. sp. *tritici* (Supplementary Tables S3 and S4). Twenty SP-SNP markers were identified for 10 avirulence loci, including three markers for *AvYr1*, one for *AvYr7*, 12 for *AvYr8*, 18 for *AvYr9*, nine for *AvYr27*, 13 for *AvYr43*, three for *AvYr44*, one for *AvYrExp2*, one for *AvYrSP*, and three for *AvYr76* (Table 5). Of the 20 SP-SNPs, PST-SNP-399 significantly associated only to *AvYr9*, and PST-SNP-56 only to *AvYr76*; all 18 other loci were significantly associated to two to eight avirulence loci. For example, PST-SNP-387 was significant associated to *AvYr1*, *AvYr8*, *AvYr9*, *AvYr27*, *AvYr43*, *AvYr44*, *AvYrSP*, and *AvYr76*; PST-SNP-415 to *AvYr1*, *AvYr8*, *AvYr9*, *AvYr27*, *AvYr43*, and *AvYr44*; and PST-SNP-363 was associated to *AvYr1*, *AvYr8*, *AvYr9*, and *AvYr43*. Only two SSR markers were significantly associated to five avirulence loci, Pstp006 to *AvYr1* and Pstp003 to *AvYr8*, *AvYr9*, *AvYr27*, and *AvYr43* (Table 6). The same markers associated to multiple avirulence loci indicate either that the avirulence loci are physically linked in the genome or that the combinations of avirulence/virulence phenotypes have been maintained in the *P. striiformis* f. sp. *tritici* populations through clonal reproduction.

DISCUSSION

Studies on virulent races have demonstrated that with added virulence factors, the post-2000 *P. striiformis* f. sp. *tritici* populations have more races, and that each race contains more virulence factors on average (Chen 2005; Chen et al. 2010; Liu et al. 2017; Wan and Chen 2014; Wan et al. 2016). The molecular marker results from this study were consistent with the previous virulence studies, and showed that the U.S. *P. striiformis* f. sp. *tritici* population has increased diversity. Here, we showed that the 2000-to-2009 *P. striiformis* f. sp. *tritici* populations had a higher heterozygosity and were more diverse than the prior-to-2000 populations. Moreover, the 2000-to-2009 *P. striiformis* f. sp. *tritici* populations were genetically distant from the prior-to-2000 populations. This study is the first to characterize the *P. striiformis* f. sp. *tritici* population in the United States by molecularly genotyping a long-term collection, and reveals that the pathogen has continually evolved over the past 40 years, with one of the major changes occurring around the year 2000.

The 2000-to-2009 *P. striiformis* f. sp. *tritici* populations generally showed higher genotypic diversity and heterozygosity (gene diversity) than the prior-to-2000 populations. Because *P. striiformis* f. sp. *tritici* is a dikaryotic fungus, single isolates can contain high variation between the two nuclei (Xia et al. 2017; Zheng et al. 2013). Focusing on a U.S. population during a short period from 2011 to 2012, Xia et al. (2016a) observed a high heterozygosity—further confirming the general inference of high heterozygosity of post-2000 *P. striiformis* f. sp. *tritici* populations. The high heterozygous populations of the post-2000 period also had a high genotypic diversity. Here, we propose two possible explanations for the high heterozygosity of the post-2000 *P. striiformis* f. sp. *tritici* populations in the United States:

First, that the high heterozygosity might have originated with the population introduced around the year 2000 and that the highly heterozygous population could be from the possible center-of-diversity region, e.g., the Transcaucasia (Stubbs 1985) or near-Himalayan regions (Ali et al. 2014).

Second, that the high heterozygosity of the post-2000 *P. striiformis* f. sp. *tritici* populations in the United States might have resulted from somatic hybridization between cereal stripe rust fungi and grass stripe rust fungi, as different formae speciales of *P. striiformis* could have coexisted in grasses (Cheng et al. 2016). To possibly support this, somatic hybridization between different formae speciales of *P. striiformis* has been confirmed in experiments under controlled conditions (Lei et al. 2017).

In addition to heterozygosity, introduction of pathogens, and somatic recombination, several other evolutionary forces may also be responsible for the high diversity and rapid change of *P. striiformis* f. sp. *tritici* populations. Mutation has been proposed to contribute to the global *P. striiformis* f. sp. *tritici* population diversity (Hovmøller et al. 2011). Many of races with similar virulence patterns in the United States were likely generated by mutations (Liu et al. 2017). Using ethyl methanesulfonate mutagenesis, our group generated 33 mutants from a least-virulent race of *P. striiformis* f. sp. *tritici* and characterized them as 24 races that included 19 new races and 21 MLGs (Li et al. 2019b). Genome sequence comparison of the mutant isolates with the progenitor isolate revealed a large number of mutation sites along the genome (Li et al. 2020). Undoubtedly, many of the similar MLGs identified in this study were created by mutation. However, the low mutation rate (Hovmøller and Justesen 2007; Xia et al. 2018) made it less possible to cause dramatic changes in a relatively short time, especially for the sudden change detected around the year 2000.

The overall *rBarD* value and its *P* value (Table 1) supported clonal reproduction of *P. striiformis* f. sp. *tritici* in the United States. Furthermore, this study determined that the *P. striiformis* f. sp. *tritici* populations of all the periods were asexually reproduced, which is consistent with the results of molecular characterization of natural *P. striiformis* f. sp. *tritici* collections in the Pacific Northwest in 2010 (Cheng and Chen 2014). Although the analysis of statistics could not rule out a low frequency of sexual reproduction, the possibility of sexual recombination in generating the high diversity in the post-2000 *P. striiformis* f. sp. *tritici* populations in the United States can be excluded; our previous studies demonstrated that *P. striiformis* f. sp. *tritici* is incapable of infecting barberry plants based on teliospore degradation during the winter season and the mismatch of barberry phenology found in the most frequent epidemic region of stripe rust in the United States, the Pacific Northwest (Chen 2005, 2014; Cheng and Chen 2014; Line 2002; Wang and Chen 2015; Wang et al. 2015).

As discussed above, the possible explanation for the rapid changes of *P. striiformis* f. sp. *tritici* populations around 2000 might be

TABLE 4. Correlation coefficients between molecular data and virulence data of *Puccinia striiformis* f. sp. *tritici* populations[a]

Period	Data 1	Data 2	Correlation coefficient	*P*
Prior 2000	SSRs	Virulence	0.274	0.205
(1968 to 1999)	SP-SNP	Virulence	0.331	0.123
	SSRs	SP-SNPs	0.838	<0.0001
Post 2000	SSRs	Virulence	0.424	0.222
(2000 to 2009)	SP-SNPs	Virulence	0.746	0.013
	SSRs	SP-SNPs	0.374	0.287
Overall	SSRs	Virulence	0.275	0.121
(1968 to 2009)	SP-SNPs	Virulence	0.356	0.042
	SSRs	SP-SNPs	0.813	<0.0001

[a] SSRs = simple sequence repeat markers; SP-SNPs = secreted protein gene-derived single nucleotide polymorphism markers.

TABLE 5. Secreted protein gene-derived single nucleotide polymorphism markers (SP-SNPs) significantly associated to avirulence (*Av*) genes identified through correlation analysis[a]

Av gene	SP-SNP	*P* value	MAF	R^2 (model without SNP)	R^2 (model with SNP)	FDR-adjusted *P* value	Effect
AvYr1	PST-SNP-387	4.15E-05	0.334	0.032	0.051	0.002029	−0.674
	PST-SNP-363	5.31E-05	0.386	0.032	0.050	0.002029	−0.470
	PST-SNP-415	6.76E-05	0.488	0.032	0.050	0.002029	0.224
AvYr7	PST-SNP-396	1.13E-05	0.267	0.082	0.103	0.001018	−0.120
AvYr8	PST-SNP-387	9.78E-11	0.334	0.234	0.272	8.80E-09	0.185
	PST-SNP-23	8.16E-08	0.097	0.234	0.260	3.67E-06	0.189
	PST-SNP-417	1.94E-07	0.444	0.234	0.258	5.83E-06	0.366
	PST-SNP-363	7.85E-07	0.386	0.234	0.256	1.77E-05	−0.242
	PST-SNP-415	1.37E-06	0.488	0.234	0.255	2.46E-05	0.343
	PST-SNP-371	1.22E-05	0.087	0.234	0.251	0.000182	−0.610
	PST-SNP-278	1.95E-05	0.186	0.234	0.250	0.000251	−0.925
	PST-SNP-48	3.31E-05	0.498	0.234	0.249	0.000335	0.273
	PST-SNP-396	3.35E-05	0.267	0.234	0.249	0.000335	1.066
	PST-SNP-432	3.98E-05	0.493	0.234	0.249	0.000358	−0.282
	PST-SNP-64	5.01E-05	0.199	0.234	0.248	0.00041	0.336
	PST-SNP-398	6.07E-05	0.400	0.234	0.248	0.000455	1.065
AvYr9	PST-SNP-387	2.38E-15	0.334	0.246	0.302	2.15E-13	−0.108
	PST-SNP-64	1.78E-11	0.199	0.246	0.286	8.01E-10	0.298
	PST-SNP-415	9.65E-11	0.488	0.246	0.283	2.90E-09	0.509
	PST-SNP-398	4.54E-10	0.400	0.246	0.280	8.70E-09	1.273
	PST-SNP-23	4.84E-10	0.097	0.246	0.280	8.70E-09	0.486
	PST-SNP-425	1.67E-08	0.494	0.246	0.274	2.51E-07	1.329
	PST-SNP-363	3.91E-08	0.386	0.246	0.272	5.03E-07	−0.310
	PST-SNP-432	1.52E-07	0.493	0.246	0.270	1.71E-06	−0.418
	PST-SNP-417	2.96E-07	0.444	0.246	0.269	2.96E-06	0.359
	PST-SNP-278	7.46E-07	0.186	0.246	0.267	6.72E-06	−1.152
	PST-SNP-436	1.72E-06	0.236	0.246	0.266	1.41E-05	−0.043
	PST-SNP-371	4.18E-06	0.087	0.246	0.264	3.13E-05	−0.658
	PST-SNP-13	4.90E-06	0.412	0.246	0.264	3.39E-05	0.526
	PST-SNP-15	1.19E-05	0.103	0.246	0.263	7.66E-05	0.320
	PST-SNP-48	2.62E-05	0.498	0.246	0.261	0.000157	0.786
	PST-SNP-399	6.74E-05	0.111	0.246	0.260	0.000379	1.308
	PST-SNP-429	8.83E-05	0.176	0.246	0.259	0.00046	0.132
	PST-SNP-17	9.19E-05	0.468	0.246	0.259	0.00046	−0.102
AvYr27	PST-SNP-387	1.65E-08	0.334	0.093	0.126	1.49E-06	−0.209
	PST-SNP-415	1.22E-06	0.488	0.093	0.118	5.17E-05	0.469
	PST-SNP-432	1.72E-06	0.493	0.093	0.117	5.17E-05	−0.072
	PST-SNP-425	4.98E-06	0.494	0.093	0.115	0.000112	1.321
	PST-SNP-13	1.56E-05	0.412	0.093	0.112	0.000281	0.644
	PST-SNP-371	2.76E-05	0.087	0.093	0.111	0.000415	−0.729
	PST-SNP-396	5.14E-05	0.267	0.093	0.110	0.000661	0.175
	PST-SNP-48	6.26E-05	0.498	0.093	0.110	0.000704	0.637
	PST-SNP-17	8.36E-05	0.468	0.093	0.109	0.000836	0.187
AvYr43	PST-SNP-387	3.05E-09	0.334	0.169	0.203	2.74E-07	0.558
	PST-SNP-415	3.11E-08	0.488	0.169	0.199	1.04E-06	0.116
	PST-SNP-23	3.46E-08	0.097	0.169	0.198	1.04E-06	0.430
	PST-SNP-417	2.31E-07	0.444	0.169	0.195	5.20E-06	0.432
	PST-SNP-436	2.03E-06	0.236	0.169	0.191	3.66E-05	−0.106
	PST-SNP-48	5.10E-06	0.498	0.169	0.189	7.65E-05	0.299
	PST-SNP-432	1.28E-05	0.493	0.169	0.187	0.000141	−0.496
	PST-SNP-371	1.31E-05	0.087	0.169	0.187	0.000141	−0.493
	PST-SNP-363	1.41E-05	0.386	0.169	0.187	0.000141	0.179
	PST-SNP-15	3.56E-05	0.1032	0.169	0.185	0.00032	−0.191
	PST-SNP-278	4.03E-05	0.186	0.169	0.185	0.00033	−0.878
	PST-SNP-398	4.51E-05	0.400	0.169	0.185	0.000338	0.923
	PST-SNP-429	9.90E-05	0.176	0.169	0.183	0.000685	−0.004
AvYr44	PST-SNP-387	1.45E-06	0.334	0.140	0.163	0.000111	−0.180
	PST-SNP-48	2.46E-06	0.498	0.140	0.162	0.000111	0.108
	PST-SNP-415	1.48E-05	0.488	0.140	0.158	0.000443	−0.139
AvYrExp2	PST-SNP-48	2.90E-06	0.498	0.098	0.121	0.000261	0.208
AvYrSP	PST-SNP-387	8.03E-05	0.334	0.016	0.033	0.007228	−0.230
AvYr76	PST-SNP-425	6.42E-05	0.494	0.036	0.054	0.002074	1.055
	PST-SNP-387	6.69E-05	0.334	0.036	0.054	0.002074	−0.443
	PST-SNP-56	6.91E-05	0.051	0.036	0.054	0.002074	−0.258

[a] MAF = minor allele frequency; FDR = false discovery rate.

somatic recombination and exotic introduction. Somatic recombination can be a possible explanation for the genotypes that appear to be hybrids or recombinants of previously existing genotypes observed in this study (data not shown). Somatic recombination has regained attention for rust research. In our studies of coinoculation of susceptible wheat plants with different combinations of *P. striiformis* f. sp. *tritici* isolates representing distinct virulence patterns under controlled conditions, we generated >80 isolates different from their "parental" isolates by either virulence phenotyping or molecular genotyping (Lei et al. 2017). Recombinant isolates were also successfully generated between *P. striiformis* f. sp. *tritici* and *P. striiformis* f. sp. *hordei* (the barley stripe rust pathogen) isolates. A 2019 study on the stem rust fungus provides evidence that the Ug99 race group arose by somatic hybridization and nuclear exchange between dikaryons (Li et al. 2019a). Their findings through whole-genome sequencing and haploid genome assembling indicate that nuclear exchange between dikaryotes can generate genetic diversity and facilitate the emergence of new lineages in asexual fungal populations. In this study, we detected few isolates of MG3 pathogen in the prior-to-2000 population. The presence of MG3 pathogen before 2000 indicates somatic recombination events between preexisting isolates and introduced isolates. This hypothesis is supported by some post-2000 races of *P. striiformis* f. sp. *tritici* with combinations of virulence genes from the *Yr8* and *Yr9* virulent races that were first detected around 2000 and those of some early races with virulence to resistance genes *Yr10* and *Yr76* (Liu et al. 2017; Wan and Chen 2014; Wan et al. 2016). The whole-genome sequencing approach can be used to test the hypothesis.

The introduction of exotic populations, followed by adaptation, could establish unique genetic lineages. Such an invasion may result in the population shift, as the case of *PstS2* in the Middle East and North Africa where newly introduced populations were strongly divergent from the established local population given in the literature (Bahri et al. 2009; Thach et al. 2016). Walter et al. (2016) evaluated the presence of aggressive *PstS1*/*PstS2* strains in the worldwide collection and observed that five out of all 11 U.S. isolates had the multilocus genotype of *PstS1*, but without the presence of *PstS2*. Coincidentally, the *PstS1* strain was first reported in the United States in 2000 (Hovmøller and Justesen 2007; Walter et al. 2016). The MG3 pathogen identified in 1998-to-2009 U.S. populations in this study is likely the *PstS1* group, but this needs to be confirmed, and its origin and migration process will need to be determined in further studies. Based on the sudden changes in the number of private MLG pathogens (Table 1), introductions of *P. striiformis* f. sp. *tritici* genotypes occurred around 1983 and 2000; this is consistent with the sudden increases in the number of races (Liu et al. 2017). However, origins of these possible introductions could not be determined in this study.

No matter where and how the new population originated, this study suggested a clear coexistence of newly introduced population with the older *P. striiformis* f. sp. *tritici* populations in the United States. As illustrated in Figure 2B, the 2000-to-2009 isolates circled by the broken curves are clustered into separate branches, while the other 2000-to-2009 isolates have relatively close genetic distances with the prior-to-2000 isolates in MG3 based on SSR markers. This demonstrates that the established *P. striiformis* f. sp. *tritici* populations in the United States have not been completely replaced by new invasions but are the result of a mixture of newly introduced and older populations. Such a trend was consistent with the virulence change in the historical *P. striiformis* f. sp. *tritici* populations in the United States (Liu et al. 2017) and other countries (de Vallavieille-Pope et al. 2012), but is in contrast to some local wheat-growing regions where gradual replacement of the preexisting population by exotic races was observed (Hovmøller et al. 2016). More extensive sampling, both spatially and temporally, and more molecular markers are needed to evaluate the contribution of new invasive races/strains to the population dynamics of *P. striiformis* f. sp. *tritici* worldwide.

The molecular genotypes from this study enable us to infer the evolutionary processes of *P. striiformis* f. sp. *tritici* populations in the United States. In Liu et al. (2017), in which the same set of *P. striiformis* f. sp. *tritici* populations was used for virulence phenotyping using 18 *Yr* single-gene differentials, we identified 171 races. Some of the races disappeared soon after the emergence in the population (PSTv-44, PSTv-48, and PSTv-71), while some other races have been in existence for >40 years (PSTv-1, PSTv-18, and PSTv-19). Some isolates of a single race collected from different periods were determined to be clonally derived, as they were molecularly similar and clustered into the same genetic groups. Comparison between molecular genotypes with virulence phenotypes also suggested that the *P. striiformis* f. sp. *tritici* populations in the United States might have been subjected to convergent selection, because some isolates of same races were clustered into distinct genetic groups (Supplementary Tables S1 and S2). The correlation analysis of the molecular genotypes with the virulence phenotypes reported in Liu et al. (2017) identified 20 SP-SNP and two SSR markers associated to 10 avirulence genes. Because the SP-SNP markers were developed from secreted protein genes and many cloned avirulence genes are determined to be secreted protein genes (Xia et al. 2016a, b, 2017), some of the SP-SNP marker genes identified to be associated with avirulence genes should be studied further to validate their relationships to the avirulence genes, which may lead to the eventual identification and cloning of the avirulence genes. The finding of same molecular markers associated to different avirulence genes is consistent with the avirulence clusters or linkages reported in Xia et al. (2020) and Yuan et al. (2018). For monitoring the pathogen, Kompetitive Allele Specific PCR (KASP) markers may be developed from the avirulence-associated SP-SNP markers for monitoring both virulence phenotypes and molecular genotypes in the *P. striiformis* f. sp. *tritici* population. Compared with SNP genotyping, KASP markers are relatively cheap and easy to use, as demonstrated in our program for stripe rust resistance genes in wheat (Liu et al. 2018, 2019, 2020a,b; Mu et al. 2020). We are, as of this writing, converting SP-SNP markers to KASP markers for monitoring *P. striiformis* f. sp. *tritici* populations.

In summary, this study revealed the coexistence of newly introduced populations with established populations in the United States. Such knowledge on dynamics of the *P. striiformis* f. sp. *tritici* population documented over >40 years is useful for understanding the genetic mechanisms of the pathogen's evolution. With the dynamics of the pathogen population in mind, framing control strategies should be emphasized on developing cultivars with nonspecific and durable resistance, ones that are effective against not only local and contemporary populations but also against the global and future populations of the pathogen.

TABLE 6. Simple sequence repeat markers (SSRs) significantly associated to avirulence (*Av*) genes identified through correlation analysis[a]

Av gene	SSRs	*P* value	MAF	R^2 (model without SNPs)	R^2 (model with SNPs)	FDR-adjusted *P* values	Effect
AvYr1	Pstp006	6.41E-05	0.371	0.051	0.069	0.000898	0.803
AvYr8	Pstp003	1.46E-09	0.397	0.230	0.263	2.05E-08	1.531
AvYr9	Pstp003	8.40E-11	0.397	0.252	0.289	1.18E-09	1.754
AvYr27	Pstp003	9.47E-06	0.397	0.089	0.110	0.000133	1.285
AvYr43	Pstp003	1.65E-06	0.397	0.171	0.193	2.31E-05	1.188

[a] SNPs = single nucleotide polymorphism markers; MAF = minor allele frequency; FDR = false discovery rate.

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LITERATURE CITED

Ali, S., Gladieux, P., Leconte, M., Gautier, A., Justesen, A. F., Hovmøller, M. S., Enjalbert, J., and de Vallavieille-Pope, C. 2014. Origin, migration routes and worldwide population genetic structure of the wheat yellow rust pathogen *Puccinia striiformis* f. sp. *tritici*. PLoS Pathog 10:e1003903.

Ali, S., Rodriguez-Algaba, J., Thach, T., Sørensen, C. K., Hansen, J. G., Lassen, P., Nazari, K., Hodson, D. P., Justesen, A. F., and Hovmøller, M. S. 2017. Yellow rust epidemics worldwide were caused by pathogen races from divergent genetic lineages. Front. Plant Sci. 8:1057.

Ali, S., Sharma, S., Leconte, M., Shah, S. J. A., Duveiller, E., Enjalbert, J., and de Vallavieille-Pope, C. 2018. Low pathotype diversity in a recombinant *Puccinia striiformis* population through convergent selection at the Eastern part of Himalayan centre of diversity (Nepal). Plant Pathol. 67:810-820.

Bahri, B., Leconte, M., Ouffroukh, A., de Vallavieille-Pope, C., and Enjalbert, J. 2009. Geographic limits of a clonal population of wheat yellow rust in the Mediterranean region. Mol. Ecol. 18:4165-4179.

Bayles, R. A., Flath, K., Hovmøller, M. S., and de Vallavieille-Pope, C. 2000. Breakdown of the *Yr17* resistance to yellow rust of wheat in northern Europe. Agronomie 20:805-811.

Beddow, J. M., Pardey, P. G., Chai, Y., Hurley, T. M., Kriticos, D. J., Braun, H.-J., Park, R. F., Cuddy, W. S., and Yonow, T. 2015. Research investment implications of shifts in the global geography of wheat stripe rust. Nat. Plants 1:15132.

Brar, G. S., Ali, S., Qutob, D., Ambrose, S., Lou, K., Maclachlan, R., and Kutcher, H. R. 2018. Genome re-sequencing and simple sequence repeat markers reveal the existence of divergent lineages in the Canadian *Puccinia striiformis* f. sp. *tritici* population with extensive DNA methylation. Environ. Microbiol. 20:1498-1515.

Chen, C. Q., Zheng, W. M., Buchenauer, H., Huang, L. L., Lu, N. H., and Kang, Z. S. 2009. Isolation of microsatellite loci from expressed sequence tag library of *Puccinia striiformis* f. sp. *tritici*. Mol. Ecol. Resour. 9:236-238.

Chen, X. M. 2005. Epidemiology and control of stripe rust [*Puccinia striiformis* f. sp. *tritici*] on wheat. Can. J. Plant Pathol. 27:314-337.

Chen, X. M. 2014. Integration of cultivar resistance and fungicide application for control of wheat stripe rust. Can. J. Plant Pathol. 36:311-326.

Chen, X. M. 2020. Pathogens which threaten food security: *Puccinia striiformis*, the wheat stripe rust pathogen. Food Secur. 12:239-251.

Chen, X. M., and Kang, Z. S. 2017. Introduction: History of research, symptoms, taxonomy of the pathogen, host range, distribution, and impact of stripe rust. Pages 1-33 in: Stripe Rust. X. M. Chen and Z. S. Kang, eds. Springer, Dordrecht, Netherlands.

Chen, X. M., Line, R. F., and Leung, H. 1993. Relationship between virulence variation and DNA polymorphism in *Puccinia striiformis*. Phytopathology 83:1489-1497.

Chen, X. M., Moore, M., Milus, E. A., Long, D. L., Line, R. F., Marshall, D., and Jackson, L. 2002. Wheat stripe rust epidemics and races of *Puccinia striiformis* f. sp. *tritici* in the United States in 2000. Plant Dis. 86:39-46.

Chen, X. M., Penman, L., Wan, A. M., and Cheng, P. 2010. Virulence races of *Puccinia striiformis* f. sp. *tritici* in 2006 and 2007 and development of wheat stripe rust and distributions, dynamics, and evolutionary relationships of races from 2000 to 2007 in the United States. Can. J. Plant Pathol. 32: 315-333.

Cheng, P., and Chen, X. M. 2014. Virulence and molecular analyses support asexual reproduction of *Puccinia striiformis* f. sp. *tritici* in the U.S. Pacific Northwest. Phytopathology 104:1208-1220.

Cheng, P., Chen, X. M., and See, D. R. 2016. Grass hosts harbor more diverse isolates of Puccinia striiformis than cereal crops. Phytopathology 106: 362-371.

Cheng, P., Chen, X. M., Xu, L. S., and See, D. R. 2012. Development and characterization of expressed sequence tag-derived microsatellite markers for the wheat stripe rust fungus *Puccinia striiformis* f. sp. *tritici*. Mol. Ecol. Resour. 12:779-781.

de Vallavieille-Pope, C., Ali, S., Leconte, M., Enjalbert, J., Delos, M., and Rouzet, J. 2012. Virulence dynamics and regional structuring of *Puccinia striiformis* f. sp. *tritici* in France between 1984 and 2009. Plant Dis. 96: 131-140.

Draz, I. S. 2019. Pathotypic and molecular evolution of contemporary population of *Puccinia striiformis* f. sp. *tritici* in Egypt during 2016–2018. J. Plant Pathol. 167:26-34.

Enjalbert, J., Duan, X., Giraud, T., Vautrin, D., de Vallavieille-Pope, C., and Solignac, M. 2002. Isolation of twelve microsatellite loci, using an enrichment protocol, in the phytopathogenic fungus *Puccinia striiformis* f. sp. *tritici*. Mol. Ecol. Notes 2:563-565.

Gardes, M., and Bruns, T. D. 1993. ITS primers with enhanced specificity for basidiomycetes–application to the identification of mycorrhizae and rusts. Mol. Ecol. 2:113-118.

Grünwald, N. J., Goodwin, S. B., Milgroom, M. G., and Fry, W. E. 2003. Analysis of genotypic diversity data for populations of microorganisms. Phytopathology 93:738-746.

Hedrick, P. W. 2005. A standardized genetic differentiation measure. Evolution 59:1633-1638.

Holtz, M. D., Kumar, K., Zantinge, J. L., and Xi, K. 2014. Genetic diversity of *Puccinia striiformis* from cereals in Alberta, Canada. Plant Pathol. 63: 415-424.

Hovmøller, M. S., and Justesen, A. F. 2007. Rates of evolution of avirulence phenotypes and DNA markers in a northwest European population of *Puccinia striiformis* f. sp. *tritici*. Mol. Ecol. 16:4637-4647.

Hovmøller, M. S., Sørensen, C. K., Walter, S., and Justesen, A. F. 2011. Diversity of *Puccinia striiformis* on cereals and grasses. Annu. Rev. Phytopathol. 49:197-217.

Hovmøller, M. S., Walter, S., Bayles, R. A., Hubbard, A., Flath, K., Sommerfeldt, N., Leconte, M., Czembor, P., Rodriguez-Algaba, J., Thach, T., Hansen, J. G., Lassen, P., Justesen, A. F., Ali, S., and de Vallavieille-Pope, C. 2016. Replacement of the European wheat yellow rust population by new races from the centre of diversity in the near-Himalayan region. Plant Pathol. 65:402-411.

Hovmøller, M. S., Walter, S., and Justesen, A. F. 2010. Escalating threat of wheat rusts. Science 329:369.

Hovmøller, M. S., Yahyaoui, A. H., Milus, E. A., and Justesen, A. F. 2008. Rapid global spread of two aggressive strains of a wheat rust fungus. Mol. Ecol. 17:3818-3826.

Jombart, T. 2008. Adegenet: a R package for the multivariate analysis of genetic markers. Bioinformatics 24:1403-1405.

Jombart, T., Devillard, S., and Balloux, F. 2010. Discriminant analysis of principal components: a new method for the analysis of genetically structured populations. BMC Genet. 11:94.

Kamvar, Z. N., Tabima, J. F., and Grunwald, N. J. 2014. *Poppr*: An R package for genetic analysis of populations with clonal, partially clonal, and/or sexual reproduction. PeerJ 2:e281.

Lei, Y., Wang, M., Wan, A., Xia, C. J., See, D. R., Zhang, M., and Chen, X. M. 2017. Virulence and molecular characterization of experimental isolates of the stripe rust pathogen (Puccinia striiformis) indicate somatic recombination. Phytopathology 107:329-344.

Li, F., Upadhyaya, N. M., Sperschneider, J., Matny, O., Nguyen-Phuc, H., Mago, R., Raley, C., Miller, M. E., Silverstein, K. A. T., Henningsen, E., Hirsch, C. D., Visser, B., Pretorius, Z. A., Steffenson, B. J., Schwessinger, B., Dodds, P. N., and Figueroa, M. 2019a. Emergence of the Ug99 lineage of the wheat stem rust pathogen through somatic hybridization. Nat. Commun. 10:5068.

Li, Y. X., Wang, M. N., See, D. R., and Chen, X. M. 2019b. Ethyl-methanesulfonate mutagenesis generated diverse isolates of *Puccinia striiformis* f. sp. *tritici*. World J. Microbiol. Biotechnol. 35:28.

Li, Y. X., Xia, C. J., Wang, M. N., Yin, C. T., and Chen, X. M. 2020. Whole-genome sequencing of *Puccinia striiformis* f. sp. *tritici* mutant isolates identifies avirulence gene candidates. BMC Genomics 21:247.

Line, R. F. 2002. Stripe rust of wheat and barley in North America: a retrospective historical review. Annu. Rev. Phytopathol. 40:75-118.

Line, R. F., and Qayoum, A. 1992. Virulence, aggressiveness, evolution and distribution of races of Puccinia striiformis (the cause of stripe rust of wheat) in North America, 1968-87. U.S. Department of Agriculture Technical Bulletin No. 1788, National Technical Information Service, Springfield, VA.

Liu, L., Wang, M. N., Feng, J. Y., See, D. R., Chao, S. M., and Chen, X. M. 2018. Combination of all-stage and high-temperature adult-plant resistance QTL confers high level, durable resistance to stripe rust in winter wheat cultivar Madsen. Theor. Appl. Genet. 131:1835-1849.

Liu, L., Wang, M. N., Zhang, Z. W., See, D. R., and Chen, X. M. 2020a. Identification of stripe rust resistance loci in U.S. spring wheat cultivars and breeding lines using genome-wide association mapping and *Yr* gene markers. Plant Dis. 104:2181-2192.

Liu, L., Yuan, C. Y., Wang, M. N., See, D. R., and Chen, X. M. 2020b. Mapping quantitative trait loci for high adult-plant resistance to stripe rust in spring wheat PI 197734 using a doubled haploid population and genotyping by multiplexed sequencing. Front. Plant Sci. 11:596962.

Liu, L., Yuan, C. Y., Wang, M. N., See, D. R., Zemetra, R. S., and Chen, X. M. 2019. QTL analysis of durable stripe rust resistance in the North American winter wheat cultivar Skiles. Theor. Appl. Genet. 132:1677-1691.

Liu, T., Wan, A., Liu, D., and Chen, X. 2017. Changes of races and virulence genes in *Puccinia striiformis* f. sp. *tritici*, the wheat stripe rust pathogen, in the United States from 1968 to 2009. Plant Dis. 101:1522-1532.

Lyon, B., and Broders, K. 2017. Impact of climate change and race evolution on the epidemiology and ecology of stripe rust in central and eastern USA and Canada. Can. J. Plant Pathol. 39:385-392.

Markell, S. G., and Milus, E. A. 2008. Emergence of a novel population of *Puccinia striiformis* f. sp. *tritici* in eastern United States. Phytopathology 98:632-639.

Meirmans, P. G., and Hedrick, P. W. 2011. Assessing population structure: F_{ST} and related measures. Mol. Ecol. Resour. 11:5-18.

Mu, J. M., Liu, L., Liu, Y., Wang, M. N., See, D. R., Han, D. J., and Chen, X. M. 2020. Genome-wide association study and gene specific markers identified 51 genes or QTL for resistance to stripe rust in U.S. winter wheat cultivars and breeding lines. Front. Plant Sci. 11:998.

Murtagh, F., and Legendre, P. 2014. Ward's hierarchical agglomerative clustering method: Which algorithms implement Ward's criterion? J. Classif. 31: 274-295.

Nei, M. 1978. Estimation of average heterozygosity and genetic distance from a small number of individuals. Genetics 89:583-590.

Park, R. F., and Wellings, C. R. 2012. Somatic hybridization in the Uredinales. Annu. Rev. Phytopathol. 50:219-239.

Peakall, R., and Smouse, P. E. 2012. GENALEX 6: genetic analysis in Excel. Population genetic software for teaching and research-an update. Bioinformatics 28:2537-2539.

Sharma-Poudyal, D., Bai, Q., Wan, A. M., Wang, M. N., See, D. R., and Chen, X. M. 2020. Molecular characterization of international collections of the wheat stripe rust pathogen *Puccinia striiformis* f. sp. *tritici* reveals high diversity and intercontinental migration. Phytopathology 110:933-942.

Stoddart, J. A., and Taylor, J. F. 1988. Genotypic diversity: estimation and prediction in samples. Genetics 118:705-711.

Stubbs, R. W. 1985. Stripe rust. Pages 61-101 in: The Cereal Rusts II: Disease, Distribution, Epidemiology and Control. A. P. Roelfs and W. R. Bushnell, eds. Academic Press, Inc., New York.

Thach, T., Ali, S., de Vallavieille-Pope, C., Justesen, A. F., and Hovmøller, M. S. 2016. Worldwide population structure of the wheat rust fungus *Puccinia striiformis* in the past. Fungal Genet. Biol. 87:1-8.

Visser, B., Herselman, L., and Pretorius, Z. A. 2016. Microsatellite characterisation of South African *Puccinia striiformis* races. South Afri. J. Plant Soil 33:161-166.

Walter, S., Ali, S., Kemen, E., Nazari, K., Bahri, B. A., Enjalbert, J., Hansen, J. G., Brown, J. K. M., Sicheritz-Pontén, T., Jones, J., de Vallavieille-Pope, C., Hovmøller, M. S., and Justesen, A. F. 2016. Molecular markers for tracking the origin and worldwide distribution of invasive strains of *Puccinia striiformis*. Ecol. Evol. 6:2790-2804.

Wan, A. M., and Chen, X. M. 2014. Virulence characterization of *Puccinia striiformis* f. sp. *tritici* using a new set of *Yr* single-gene line differentials in the United States in 2010. Plant Dis. 98:1534-1542.

Wan, A. M., Chen, X. M., and Yuen, J. 2016. Races of *Puccinia striiformis* f. sp. *tritici* in the United States in 2011 and 2012 and comparison with races in 2010. Plant Dis. 100:966-975.

Wang, J., and Zhang, Z. 2018. GAPIT Version 3: An Interactive Analytical Tool for Genomic Association and Prediction. https://www.researchgate.net/publication/329829469_GAPIT_Version_3_An_Interactive_Analytical_Tool_for_Genomic_Association_and_Prediction

Wang, M. N., and Chen, X. M. 2015. Barberry does not function as an alternate host for *Puccinia graminis* f. sp. *tritici* in the U. S. Pacific Northwest due to teliospore degradation and barberry phenology. Plant Dis. 99:1500-1506.

Wang, M. N., Wan, A. M., and Chen, X. M. 2015. Barberry as alternate host is important for *Puccinia graminis* f. sp. *tritici* but not for *Puccinia striiformis* f. sp. *tritici* in the U. S. Pacific Northwest. Plant Dis. 99:1507-1516.

Wellings, C. R. 2011. Global status of stripe rust: a review of historical and current threats. Euphytica 179:129-141.

Wellings, C. R., Boyd, L. A., and Chen, X. M. 2012. Resistance to stripe rust in wheat: pathogen biology driving resistance breeding. Pages 63-83 in: Disease Resistance in Wheat. Sharma, I., ed. CAB International, Wallingford, United Kingdom.

Wellings, C. R., and McIntosh, R. A. 1990. *Puccinia striiformis* f. sp. *tritici* in Australasia: Pathogenic changes during the first 10 years. Plant Pathol. 39: 316-325.

Xia, C. J., Lei, Y., Wang, M. N., Chen, W. Q., and Chen, X. M. 2020. An avirulence gene cluster in the wheat stripe rust pathogen (*Puccinia striiformis* f. sp. *tritici*) identified through genetic mapping and whole-genome sequencing of a sexual population. MSphere 5:e00128-e20.

Xia, C. J., Wan, A. M., Wang, M. N., Jiwan, D. A., See, D. R., and Chen, X. M. 2016a. Secreted protein gene derived-single nucleotide polymorphisms (SP-SNPs) reveal population diversity and differentiation of *Puccinia striiformis* f. sp. *tritici* in the United States. Fungal Biol. 120:729-744.

Xia, C. J., Wang, M. N., Cornejo, O. E., Jiwan, D. A., See, D. R., and Chen, X. M. 2017. Secretome characterization and correlation analysis reveal putative pathogenicity mechanisms and identify candidate avirulence genes in the wheat stripe rust fungus *Puccinia striiformis* f. sp. *tritici*. Front. Microbiol. 8:2394.

Xia, C. J., Wang, M. N., Wan, A. M., Jiwan, D. A., See, D. R., and Chen, X. M. 2016b. Association analysis of SP-SNPs and avirulence genes in *Puccinia striiformis* f. sp. *tritici*, the wheat stripe rust pathogen. Am. J. Plant Sci. 7:126-137.

Xia, C. J., Wang, M. N., Yin, C. T., Cornejo, O. E., Hulbert, S. H., and Chen, X. M. 2018. Genomic insights into host adaptation between the wheat stripe rust pathogen (*Puccinia striiformis* f. sp. *tritici*) and the barley stripe rust pathogen (*Puccinia striiformis* f. sp. *hordei*). BMC Genomics 19:664.

Yuan, C. Y., Wang, M. N., Skinner, D. Z., See, D. R., Xia, C. J., Guo, X. H., and Chen, X. M. 2018. Inheritance of virulence, construction of a linkage map, and mapping of virulence genes in *Puccinia striiformis* f. sp. *tritici* by virulence and molecular characterization of a sexual population through genotyping-by-sequencing. Phytopathology 108:133-141.

Zhao, J., Wang, M. N., Chen, X. M., and Kang, Z. S. 2016. Role of alternate hosts in epidemiology and pathogen variation of cereal rusts. Annu. Rev. Phytopathol. 54:207-228.

Zheng, W. M., Huang, L. L., Huang, J. Q., Wang, X. J., Chen, X. M., Zhao, J., Guo, J., Zhuang, H., Qiu, C., Liu, J., Liu, H., Huang, X., Pei, G., Zhan, G., Tang, C., Cheng, Y., Liu, M., Zhang, J., Zhao, Z., Zhang, S., Han, Q., Han, D., Zhang, H., Zhao, J., Gao, X., Wang, J., Ni, P., Dong, W., Yang, L., Yang, H., Xu, J.-R., Zhang, G., and Kang, Z. 2013. High genome heterozygosity and endemic genetic recombination in the wheat stripe rust fungus. Nat. Commun. 4:2673.